

Luke Yamaguchi

Irvine, CA | yamaluke@g.ucla.edu | yamaluke.com | linkedin.com/in/yamaluke | US Citizen

EDUCATION

University of California, Los Angeles (UCLA)

Sep 2024 - Dec 2027

M.S. Electrical & Computer Engineering (Exceptional Student Admission Program)

Expected Dec 2027

- *Relevant Coursework:* Advanced GPU Microarchitecture (in progress), Advanced Computer Architecture (in progress)

B.S. Computer Engineering

Jun 2026

- *GPA:* 3.86/4.00

- *Relevant Coursework:* Computer Systems Architecture, Digital Circuits, Computer System Security, Operating Systems, Digital Signal Processing, Algorithms and Complexity

EXPERIENCE

Graduate ASIC/FPGA Security Engineering Intern

Los Angeles, CA

The Aerospace Corporation

Jun 2026 - Present

- Developed BBRAM anti-tamper zeroization test suite protecting cryptographic key storage on AMD Versal VEK280
 - Built Monte Carlo based fault injection emulation to evaluate robustness against radiation induced faults
 - Programmed bare-metal C firmware for RPU and Vivado/Vitis infrastructure for SoC security verification
- Implemented autonomous SoC threat detection and mitigation pipeline on VEK280 APU/PetaLinux
 - Designed LLM prompting workflow to create threat model identifying attack surfaces and mitigations
 - Developed telemetry anomaly detection using a variational autoencoder, achieving 100% accuracy in initial testing

Undergraduate Researcher

Los Angeles, CA

Secure Systems and Architectures Lab - UCLA

Oct 2025 - Jun 2026

- Researched BLE hardware security and RF device authentication using PHY-layer hardware fingerprints
- Built GNU Radio DSP pipeline to extract physical-layer features from BLE signals captured from ADALM-Pluto SDR
- Trained ML-based RF fingerprinting models on Linux remote servers
- Developed universal adversarial perturbation attacks against RF fingerprinting models and conducted over-the-air spoofing experiments using HackRF SDR

PROJECTS

Hardware Security Lead — Mini-GPU FPGA Accelerator

Mar 2026 - Jun 2026

- Built a hardware root of trust for kernel verification with golden hashes, launch gating, and program-memory locking
- Threat modeled the system and patched 5 attack vectors, including a residual memory exploit after reset
- Cut SHA-256 core area 54% (10.5k to 4.9k cells) by serializing the digest adder and removing parallel logic
- Closed timing at 75 MHz in Vivado with the secure design at 97.5% LUT utilization on Basys3 FPGA
- Prototyped a 4 tier key derivation and AES/HMAC encryption scheme, removed after synthesis exceeded LUT budget
- Designed custom protocol with CRC validated framing and CDC safe ACK/NACK handshaking to memory controller

Hardware-Enforced Authentication System

Feb 2025 - Mar 2025

- Designed and implemented multi-user authentication system in Verilog on an Artix-7 FPGA
- Built hardware-enforced role-based access control (RBAC) with admin, user, and guest privilege separation
- Implemented dynamic credential lifecycles, including creation, modification, and deletion
- Engineered fail-secure features (auto re-locking & brute-force lockout) informed by authentication threat modeling

Lead Researcher — Multi-Agent Access Control

Oct 2025 - Dec 2025

- Proposed a provenance-based access control framework using instruction-origin tainting to mitigate Confused Deputy attacks in multi-agent LLM systems
- Reduced attack success rate from 57% to 29% across 15 test cases

SKILLS

Certificate: CompTIA Security+ (Aug 2026)

Languages / HDLs: Verilog, SystemVerilog, C, C++, Python, Bash

Verification/Synthesis: Testbenches, Vivado, Yosys, Icarus Verilog

Tools/Lab: Git, PetaLinux, Docker, GDB

Languages: English, Japanese